AENL226 Power Electronics and power systems (Aug-Dec 2026)

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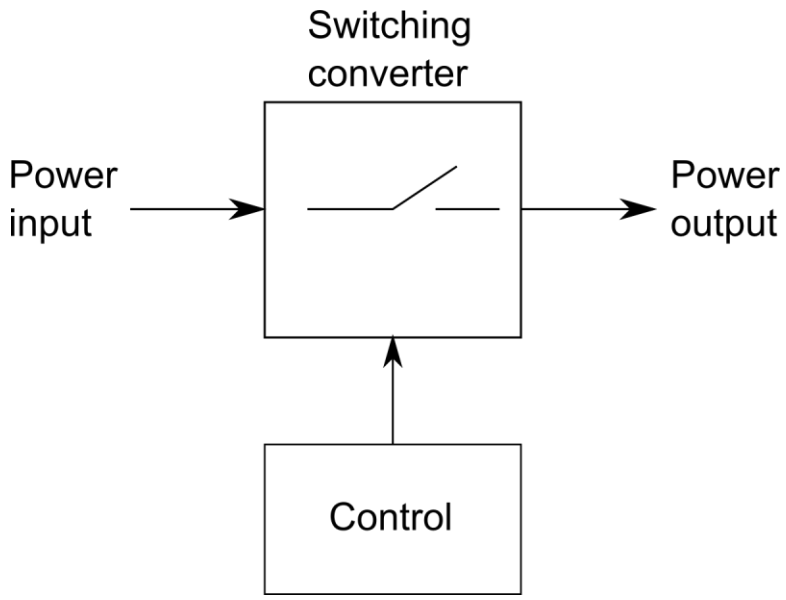
Lecture content

- Introduction to Power Electronics
- Main textbook: Issa Batarseh, Power Electronics: Circuit Analysis and Design, Springer, available in our library and Kortext
- NPTEL lectures

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What is power electronics?

- Power electronics is a subject that deals with processing of electrical power using electronic devices.
- It is an interdisciplinary subject in electrical engineering covering electronics, power and control.



Components in power electronics

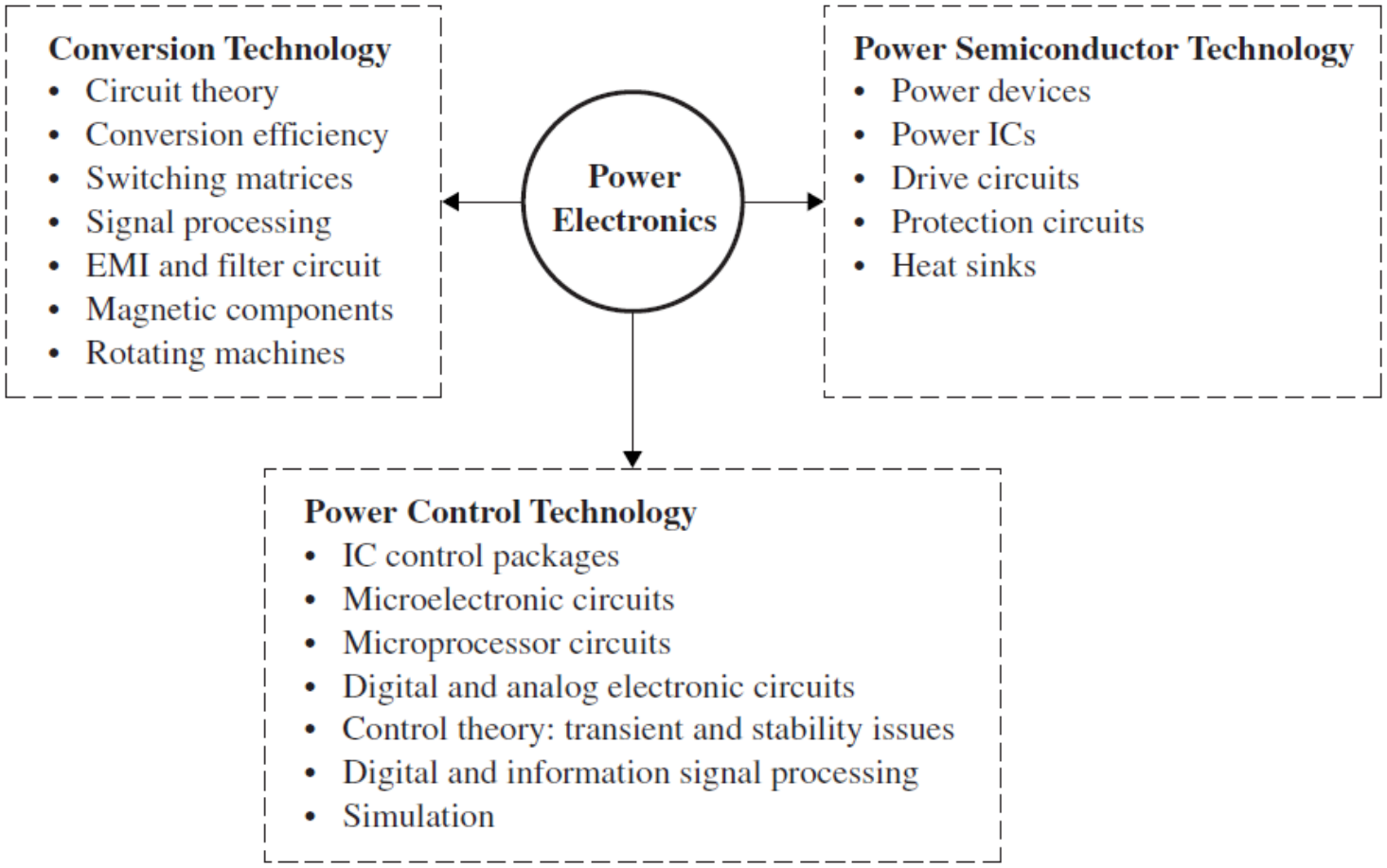


Figure 1.1 Power electronics encompasses three technologies: *power semiconductor*, *power conversion*, and *power control*.

Why are we studying this subject?

- Energy market is changing throughout the world
- Energy conversion is now mandatorily controlled by high power electronics
- Lot of prospect now and in near future

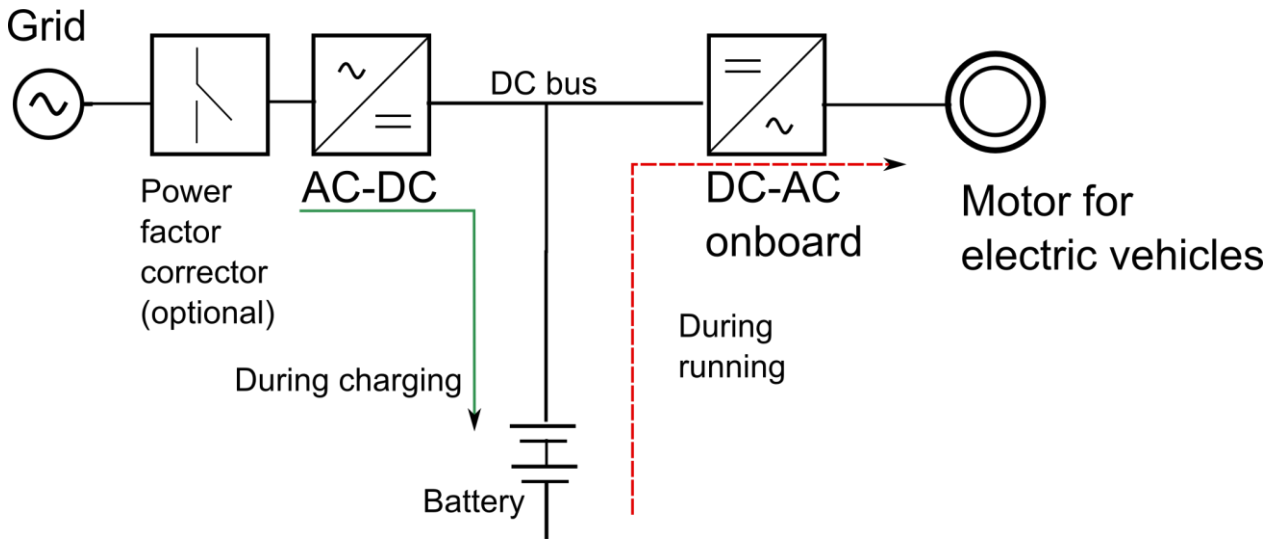
Application of power electronics

- Motor drives including electric vehicles
- Renewable energy integration (Solar, wind etc.)
- Power supplies (e.g., Chargers for EV, Uninterruptable power supplies)
- Energy efficiency improvement (e.g., heating/cooling, LED lighting)
- Utility applications (e.g., power factor correctors)
- Many more.

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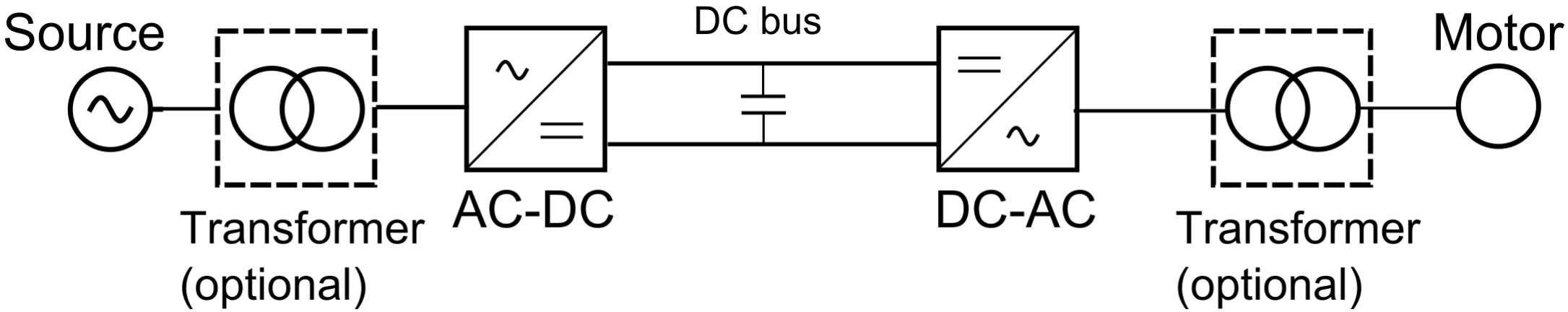
Electric vehicle

- Batteries are charged from the grid through an AC-DC rectifier.
- AC motors are controlled through DC-AC converter.



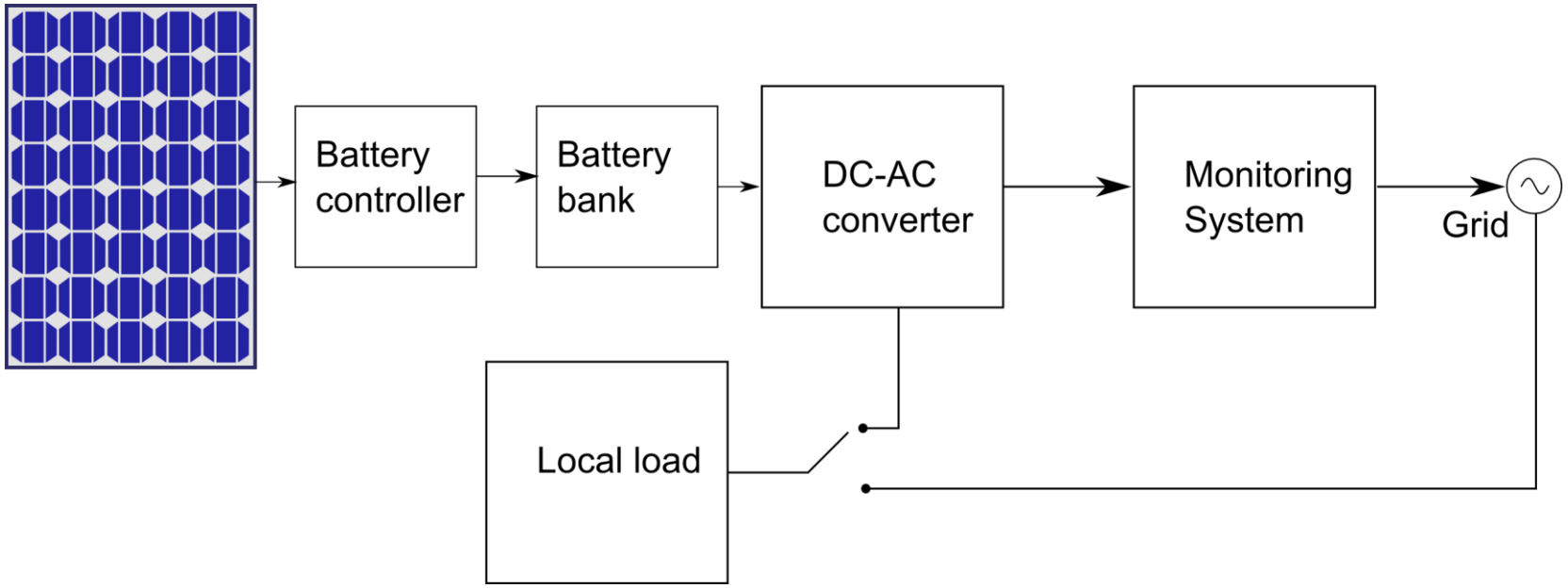
Motor drives

- Variable speed drives are required in many applications for better efficiency and better control over speed of the motor.
- Motors consume about 50% of total energy of the world.
- Power electronics play a vital role in control of motor drives.

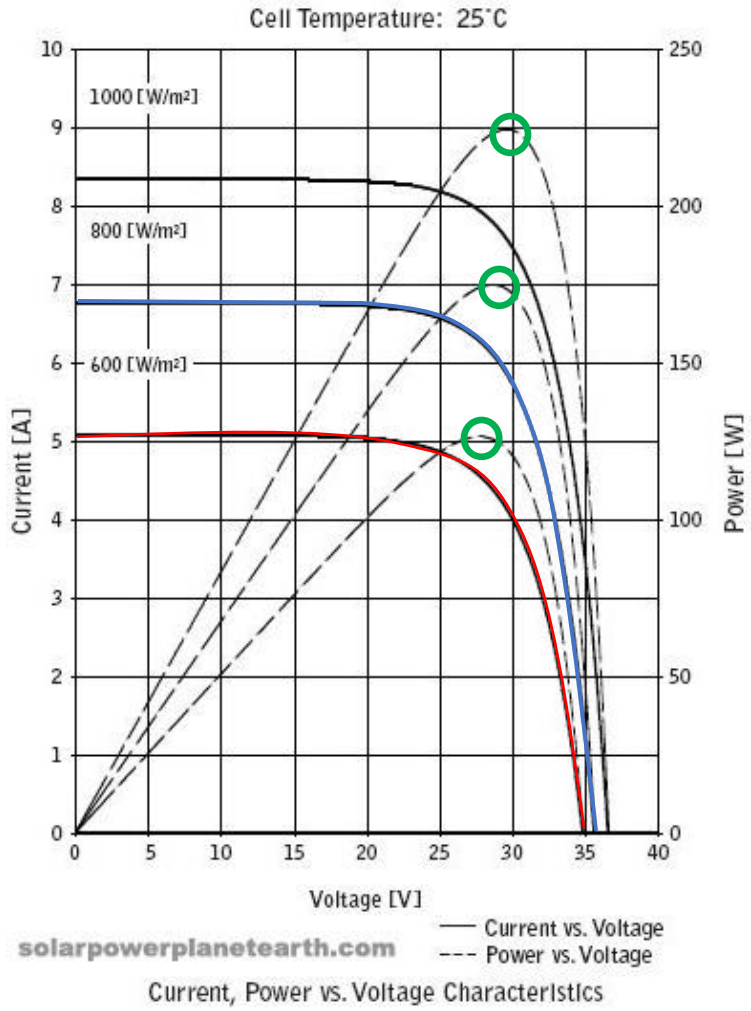


Power electronics in solar PV application

- Growth of solar energy in world
  - Current generation around 2700 TWh (as of 2026)
  - Up to 1000 GW=1TW addition every year
  - Several solar parks with power capacity excess of 2 GW is operational
- Components of PV system

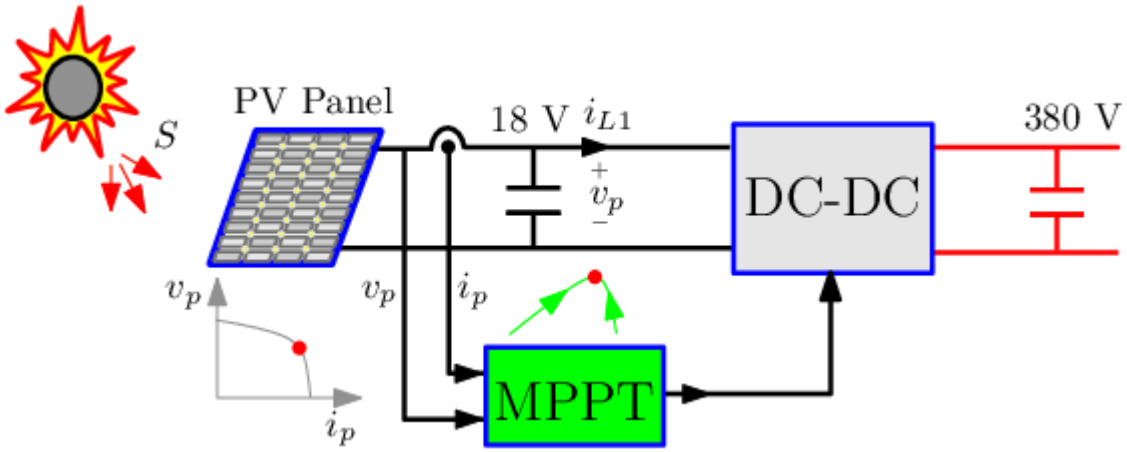


Maximum power point tracking



- When the radiation (W/m^2) changes, the I-V characteristics changes. (I: current drawn from the PV panel, V: voltage across the panel)
- Consequently, the power obtainable from the solar panel changes. It has a maximum power point.

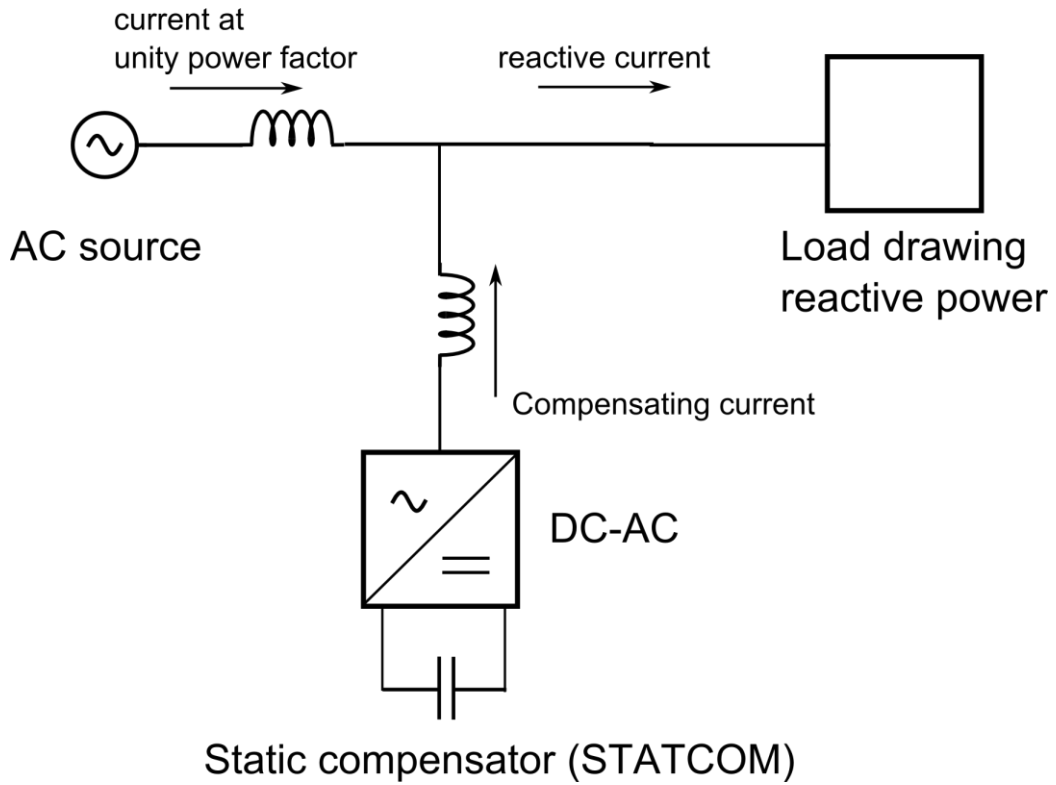
Maximum power point tracking



- In order to extract maximum power from the PV all the time, the I-V characteristics of the PV panel should be varied dynamically depending on the solar insolation and temperature.
- This dynamic variation is done through a power converter.
- This is called maximum power point tracking of a solar PV panel.

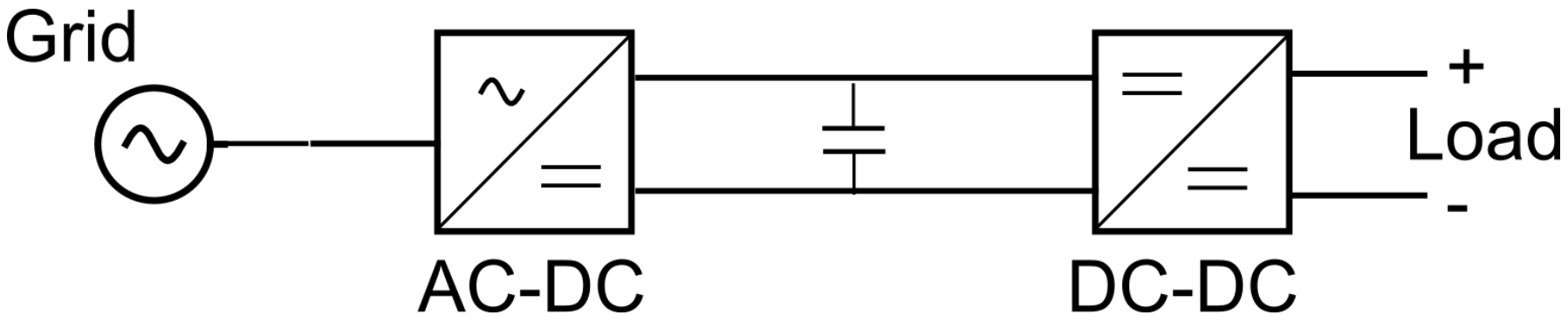
Power factor correctors

- The load e.g. a motor draws reactive current.
- To make the source supply current at unity power factor, a compensator is used.
- The compensator supplies the reactive part of the load current.



Power supplies

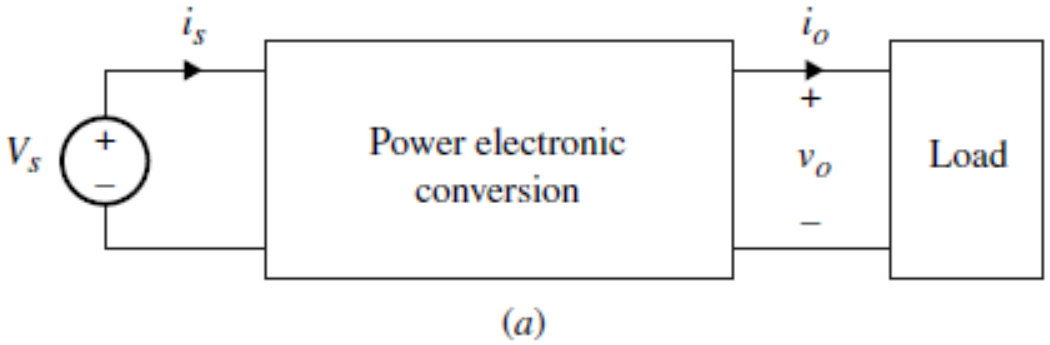
- The incoming grid supply (often single phase) is converted to DC.
- Lower values of DC voltage is obtained through a DC-DC converter.



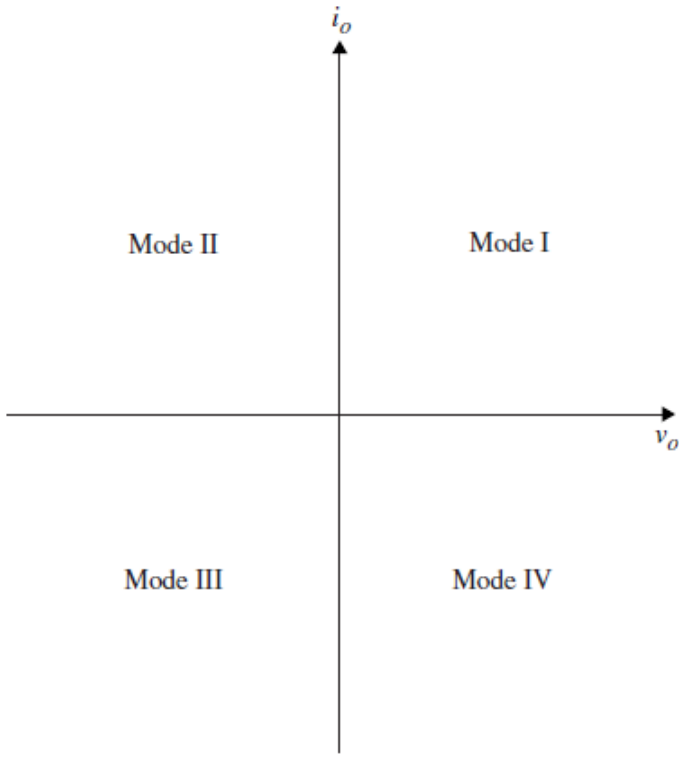
Main types of conversion circuits

- AC to DC converter called rectifier
  - Fixed AC to variable DC
- DC to AC converter called inverter
  - Fixed DC to variable AC
- DC to DC converter called dc-dc converter
  - Fixed DC to variable DC
- AC to AC converter called cycloconverter
  - Fixed AC to variable AC
  - Cycloconverters are becoming obsolete. They are getting replaced by two stage conversion: (AC to DC converter) and then (DC to AC converter).

Four quadrant operation



(a)



- Depending on the polarity of output voltage and output current, four quadrant operation is possible.
- In mode 1 and 3, load is absorbing power from source.
- In mode 2 and 4, load is delivering power back to source.